



SOC-analyst-dashboard

proof of concept

Elyes Manai, PhD ~ Arno Paumier, student

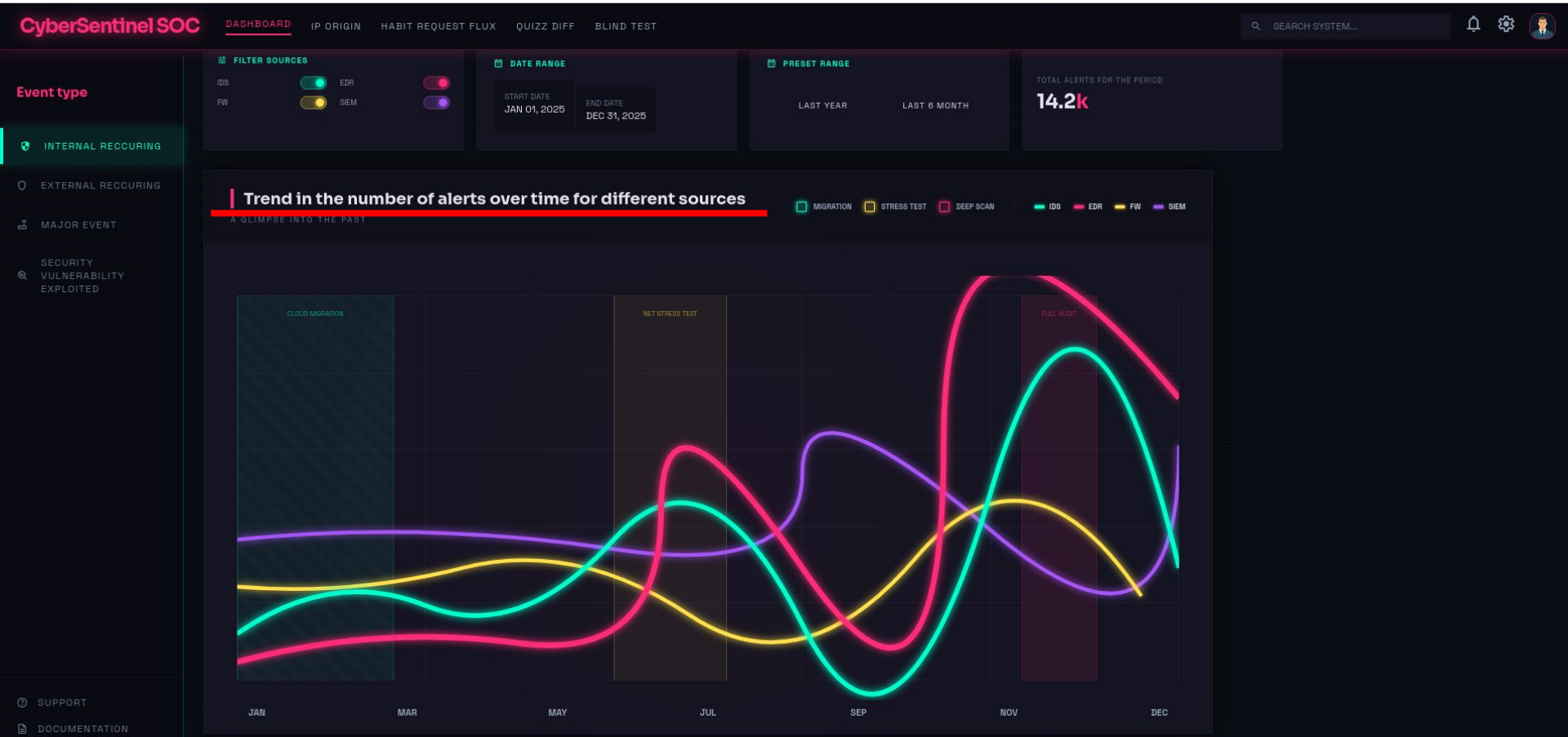
Issue

How can we help a new analyst settle into their new SOC more quickly?

Disclaimer

AI was used to generate the basic design. Some pages have different origins, which explains the noticeable differences that may exist between each design. Consistency across the pages is not entirely guaranteed

The data presented in this document is random and does not reflect any actual situation



home dashboard : graph

5



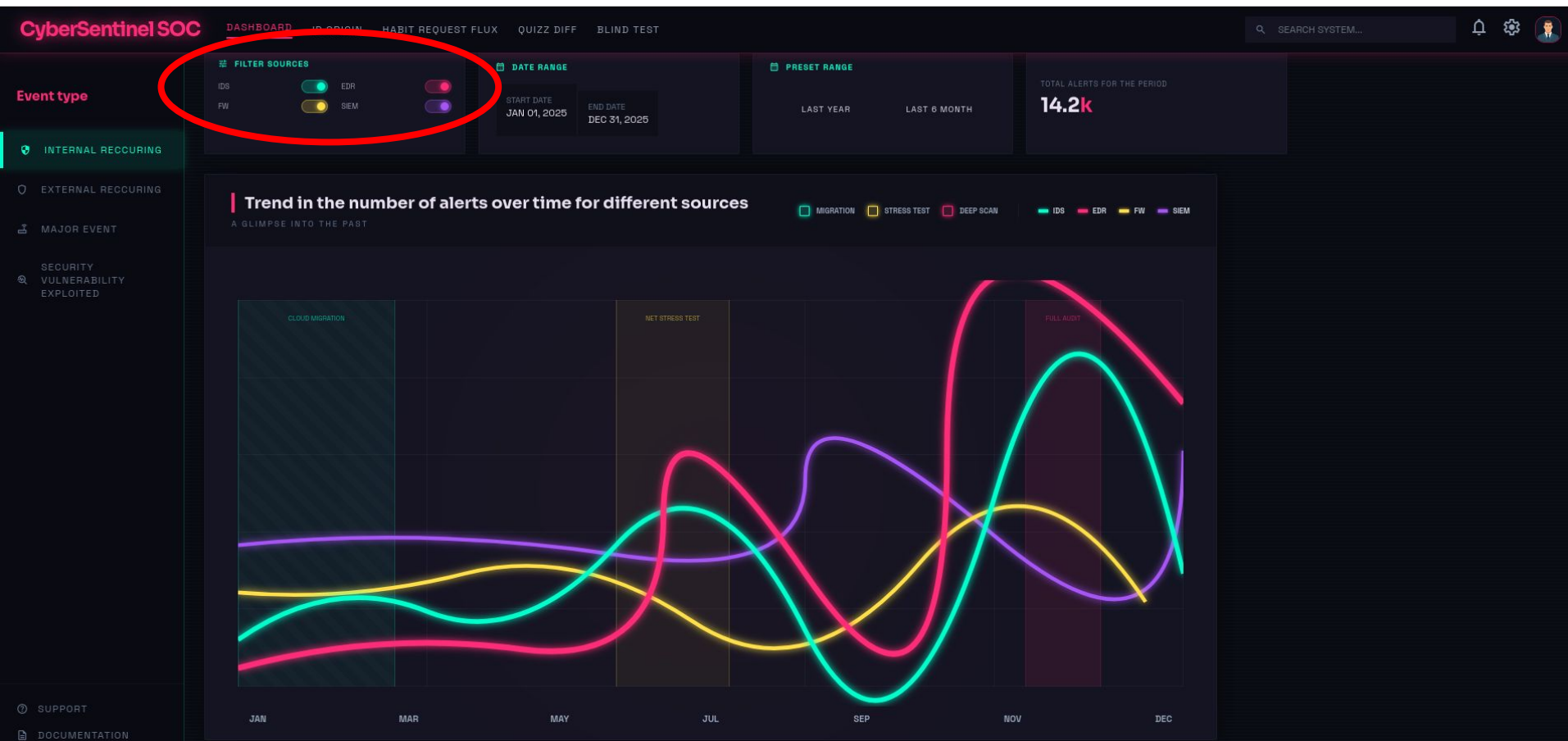
home dashboard : change displayed events

6



home dashboard : change displayed graph

7



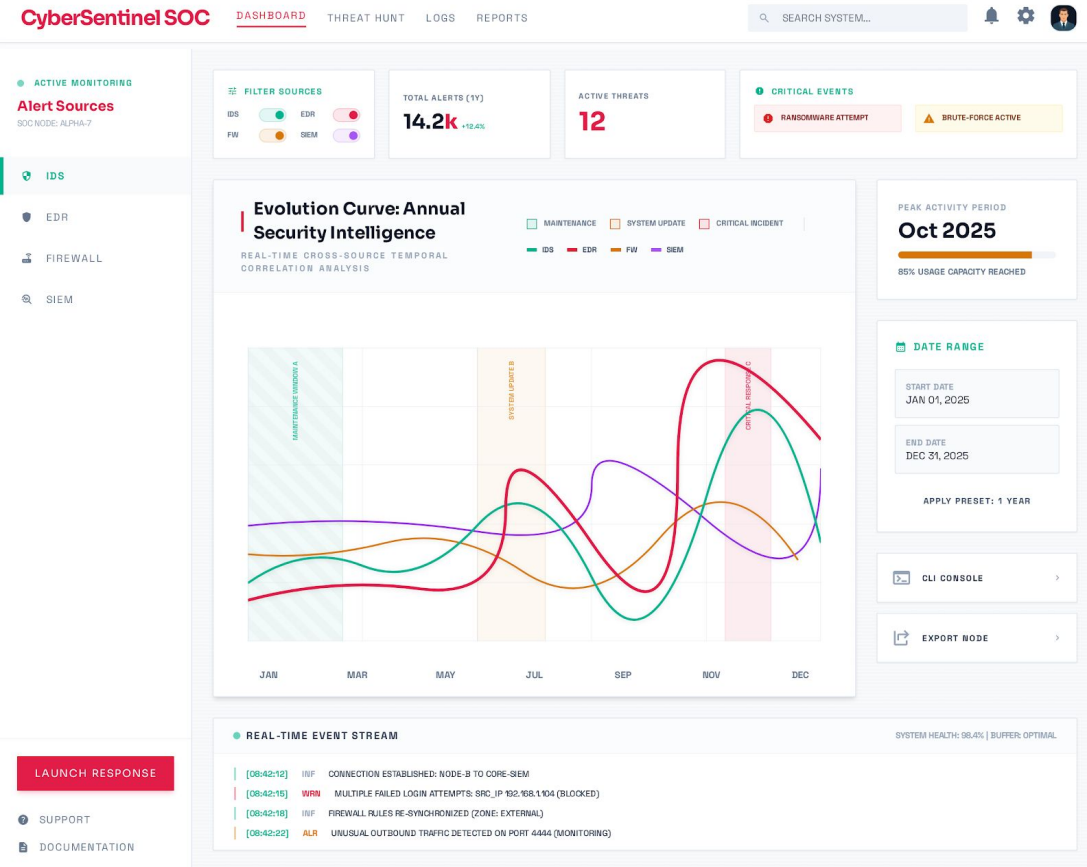
home dashboard : change range date

8



Light mode

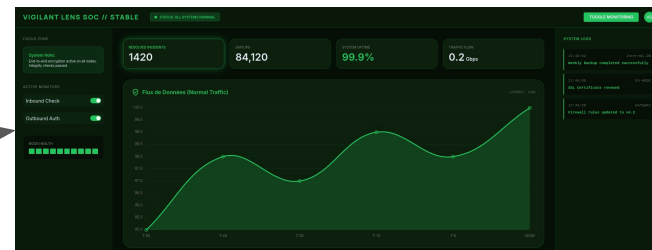
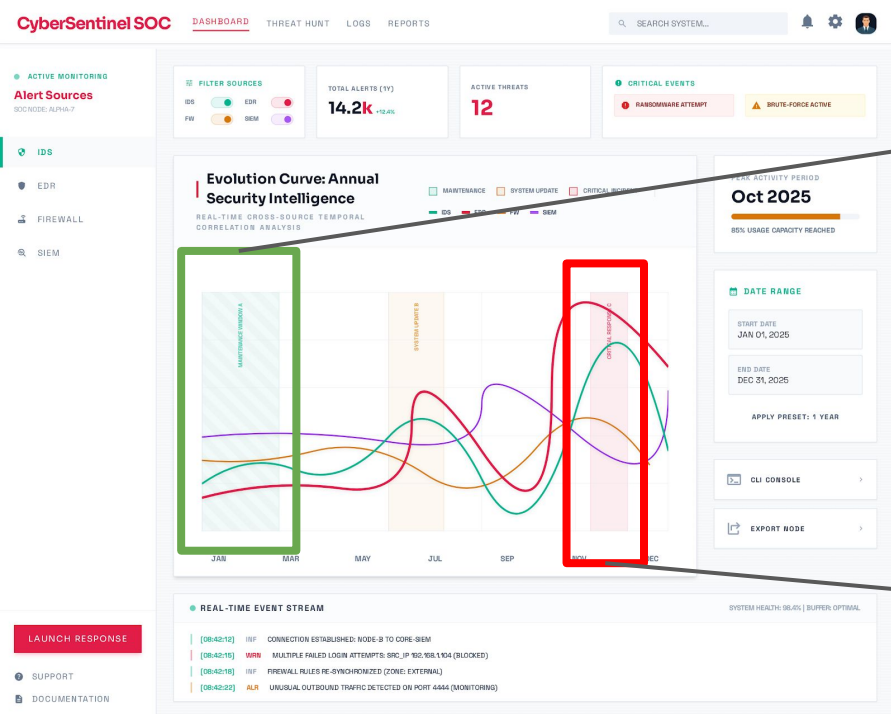
9



- There is a light mode available to protect your eyes depending on the ambient light
- One could also envisage a different layout for the modules and other information

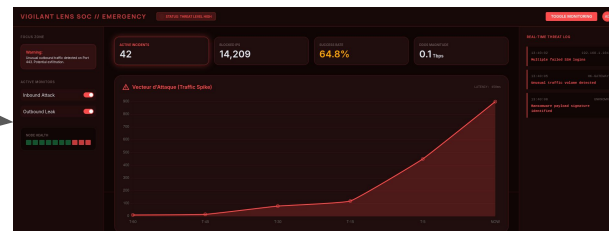
More information about events

- When you click on an event you open a new windows with more information about this event



Stable

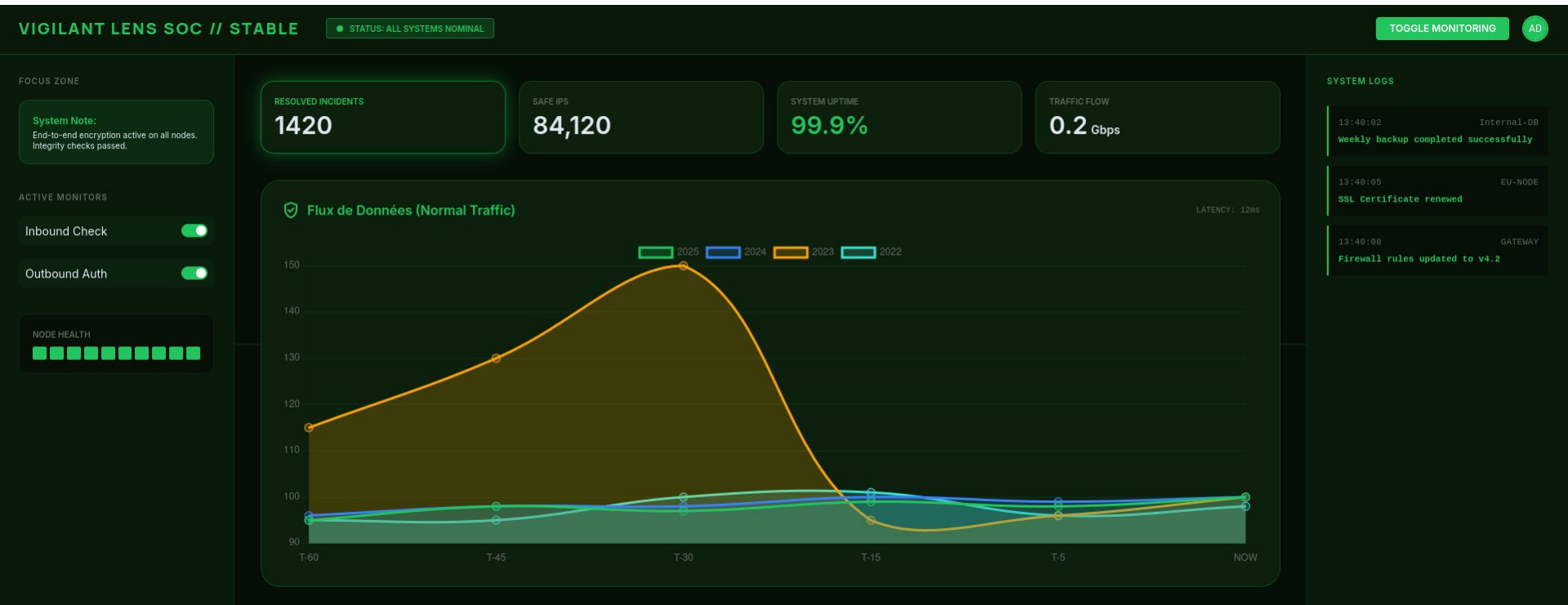
- The colour indicates the severity of the incident



Emergency

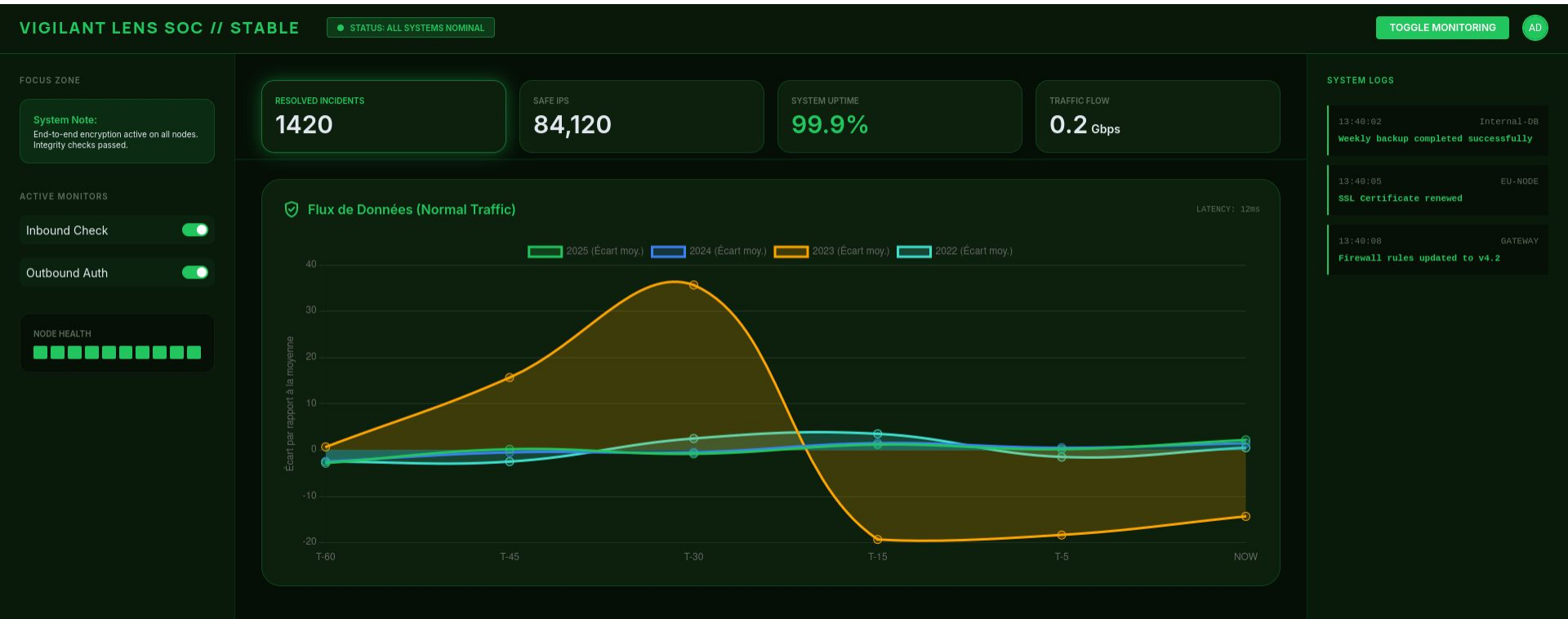
Overlapping events in the same category

This only works if the event is recurring



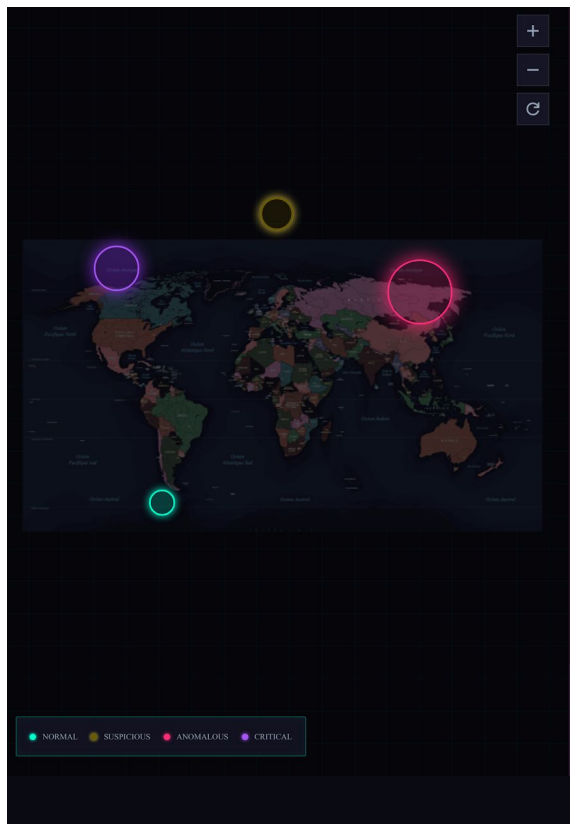
Overlapping events in the same category

You can also use a deviation from the mean

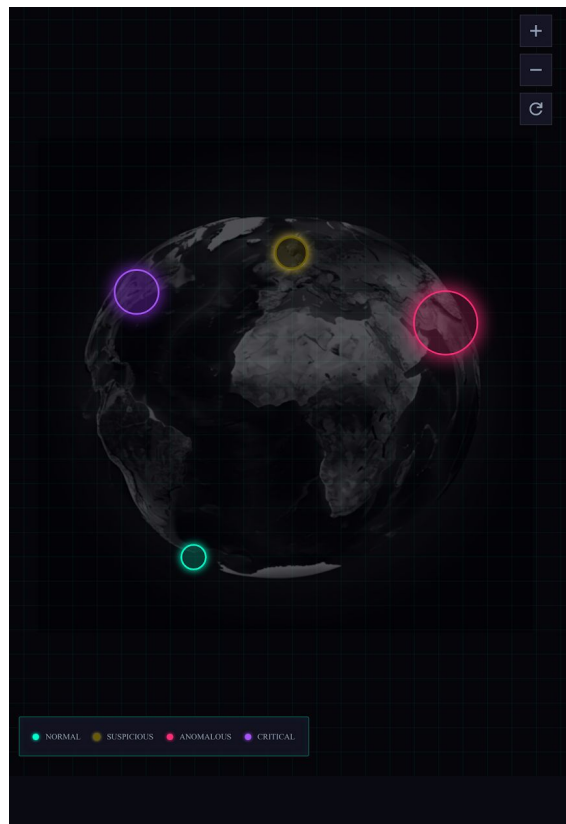


IP origin

We return to a macroscopic scale in the spatial dimension



OR



IP origin : global

14

CyberSentinel SOC

DASHBOARD

IP ORIGIN

HABIT REQUEST FLUX

QUIZZ DIFF

BLIND TEST

SEARCH SYSTEM...



Region of the world

GLOBAL

EUROPE

NORTH AMERICA

SOUTH AMERICA

ASIA

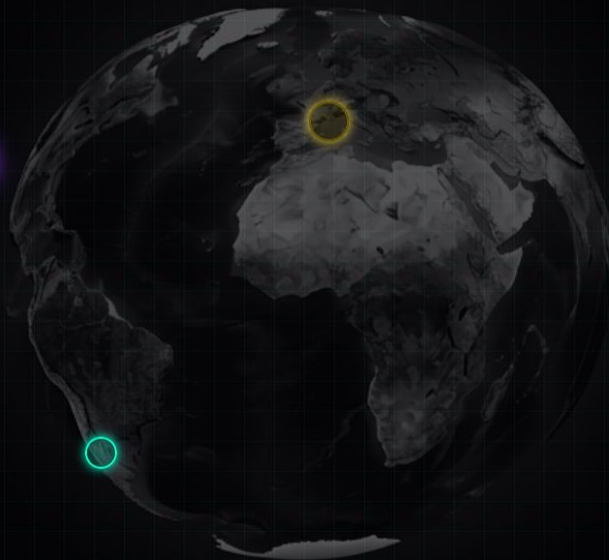
OCEANIA

AFRICA

SUPPORT

DOCUMENTATION

● NORMAL ● SUSPICIOUS ● ANOMALOUS ● CRITICAL



DATE RANGE

START DATE
JAN 01, 2025

END DATE
DEC 31, 2025

TOP THREAT ORIGINS

RUSSIAN FEDERATION

CRITICAL

REQUESTS
4,281,902

ANOMALY LEVEL

NIGERIA

CRITICAL

REQUESTS
81,902

ANOMALY LEVEL

CHINA

SUSPICIOUS

REQUESTS
2,105,443

ANOMALY LEVEL

UNKNOWN SOURCE

MODERATE

REQUESTS
892,310

ANOMALY LEVEL

AUSTRALIA

LOW

REQUESTS
457,504

ANOMALY LEVEL

SUMMARY

Normal

TOTAL REQUESTS

14.2M

TOTAL ALERTS

2,481

IP origin : Move the camera

15

CyberSentinel SOC

DASHBOARDIP ORIGINHABIT REQUEST FLUXQUIZZ DIFFBLIND TEST

SEARCH SYSTEM...

Region of the world

- GLOBAL
- EUROPE
- NORTH AMERICA
- SOUTH AMERICA
- ASIA
- OCEANIA
- AFRICA

+

-

↺

● NORMAL

● SUSPICIOUS

● ANOMALOUS

● CRITICAL

DATE RANGE

START DATE: JAN 01, 2025END DATE: DEC 31, 2025

TOP THREAT ORIGINS

RUSSIAN FEDERATION

CRITICAL

REQUESTS: 4,281,902

ANOMALY LEVEL: [red bar]

NIGERIA

CRITICAL

REQUESTS: 81,902

ANOMALY LEVEL: [red bar]

CHINA

SUSPICIOUS

REQUESTS: 2,105,443

ANOMALY LEVEL: [yellow bar]

UNKNOWN SOURCE

MODERATE

REQUESTS: 892,310

ANOMALY LEVEL: [orange bar]

AUSTRALIA

LOW

REQUESTS: 457,504

ANOMALY LEVEL: [green bar]

SUMMARY

Normal

TOTAL REQUESTS: 14.2MTOTAL ALERTS: 2,481

IP origin : View the most critical origin

16

CyberSentinel SOC

DASHBOARD

IP ORIGIN

HABIT REQUEST FLUX

QUIZZ DIFF

BLIND TEST

SEARCH SYSTEM...



Region of the world

GLOBAL

EUROPE

NORTH AMERICA

SOUTH AMERICA

ASIA

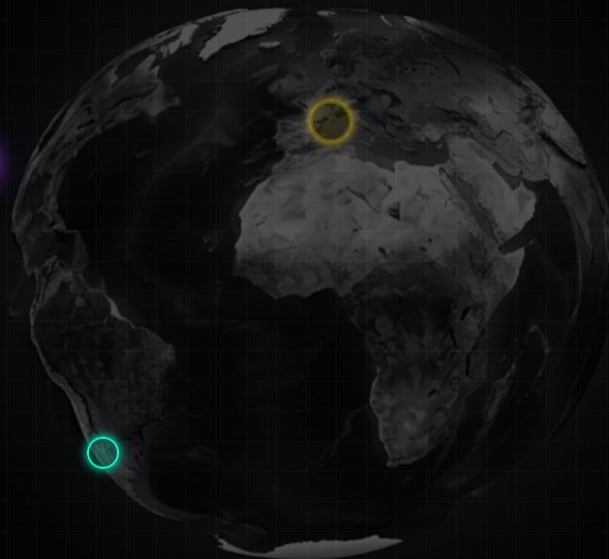
OCEANIA

AFRICA

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892,310

ANOMALY LEVEL

AUSTRALIA

LOW

REQUESTS
457,504

ANOMALY LEVEL

SUMMARY

Normal

TOTAL REQUESTS

14.2M

TOTAL ALERTS

2,481

IP origin : Zoom in on an area

17

CyberSentinel SOC

DASHBOARD

IP ORIGIN

HABIT REQUEST FLUX

QUIZZ DIFF

BLIND TEST

SEARCH SYSTEM...



Region of the world

GLOBAL

EUROPE

NORTH AMERICA

SOUTH AMERICA

ASIA

OCEANIA

AFRICA

USA_EAST_04

NORTH AMERICA

STATUS: CRITICAL BREACH

VOLUME: 1.2M req/min

ANOMALY: 92.4%

ALERTS: 14,209

DATE RANGE

START DATE
JAN 01, 2025

END DATE
DEC 31, 2025

TOP THREAT ORIGINS

RUSSIAN FEDERATION

CRITICAL

REQUESTS
4,281,902

ANOMALY LEVEL

NIGERIA

CRITICAL

REQUESTS
81,902

ANOMALY LEVEL

CHINA

SUSPICIOUS

REQUESTS
2,105,443

ANOMALY LEVEL

UNKNOWN SOURCE

MODERATE

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892,310

ANOMALY LEVEL

AUSTRALIA

LOW

REQUESTS
457,504

ANOMALY LEVEL

SUMMARY

Normal

TOTAL REQUESTS

14.2M

TOTAL ALERTS

2,481

SUPPORT

DOCUMENTATION

● NORMAL ● SUSPICIOUS ● ANOMALOUS ● CRITICAL

IP origin : change range date

18

CyberSentinel SOC

DASHBOARD

IP ORIGIN

HABIT REQUEST FLUX

QUIZZ DIFF

BLIND TEST

SEARCH SYSTEM...



Region of the world

GLOBAL

EUROPE

NORTH AMERICA

SOUTH AMERICA

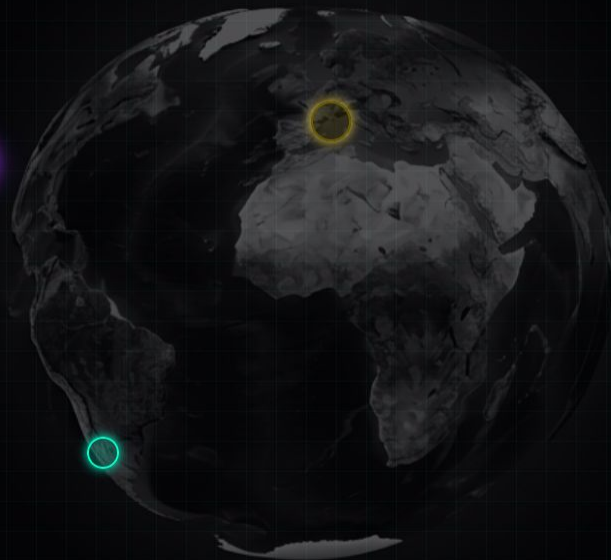
ASIA

OCEANIA

AFRICA

SUPPORT

DOCUMENTATION



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DATE RANGE

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ANOMALY LEVEL

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MODERATE

REQUESTS
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AUSTRALIA

LOW

REQUESTS
457,504

ANOMALY LEVEL

SUMMARY

Normal

TOTAL REQUESTS

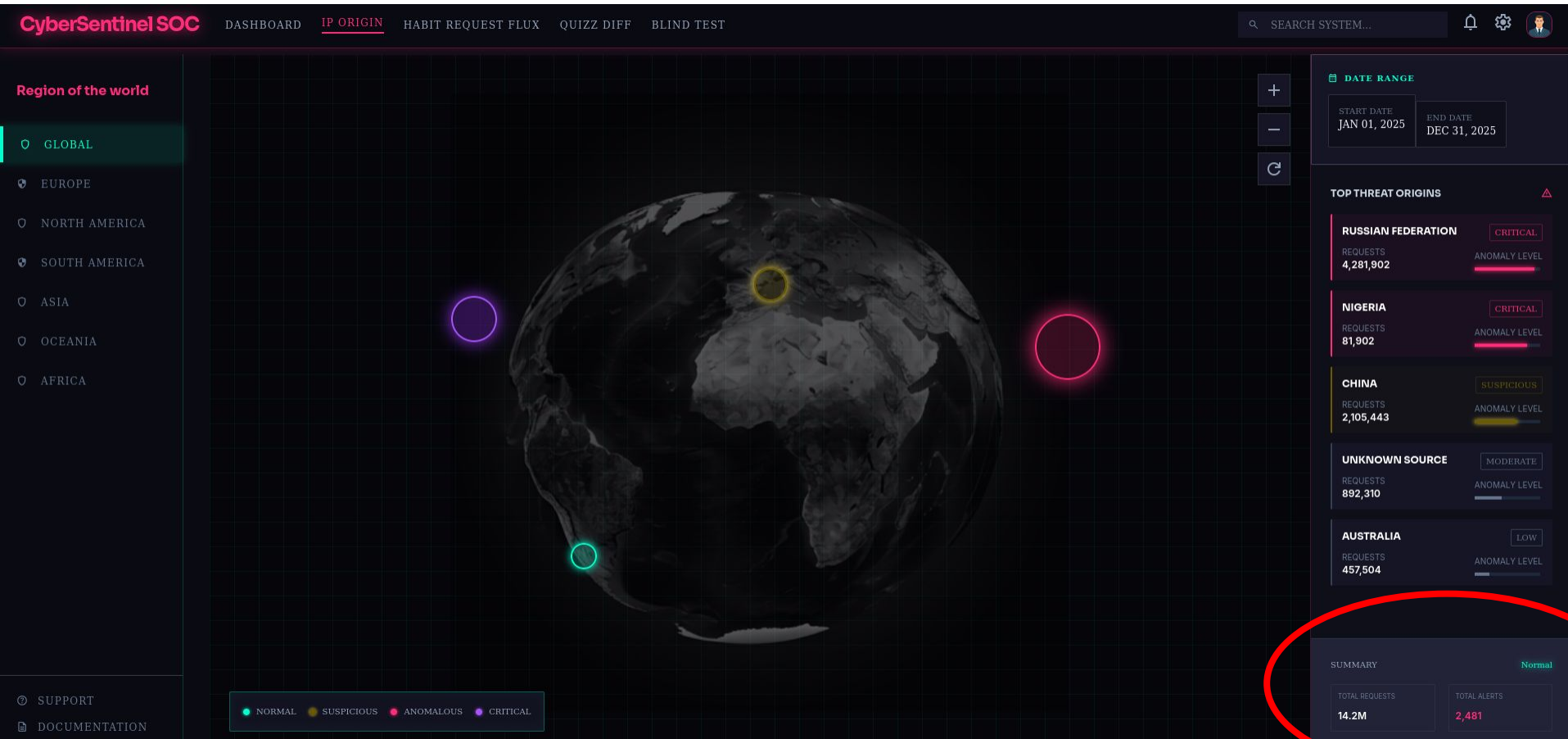
14.2M

TOTAL ALERTS

2,481

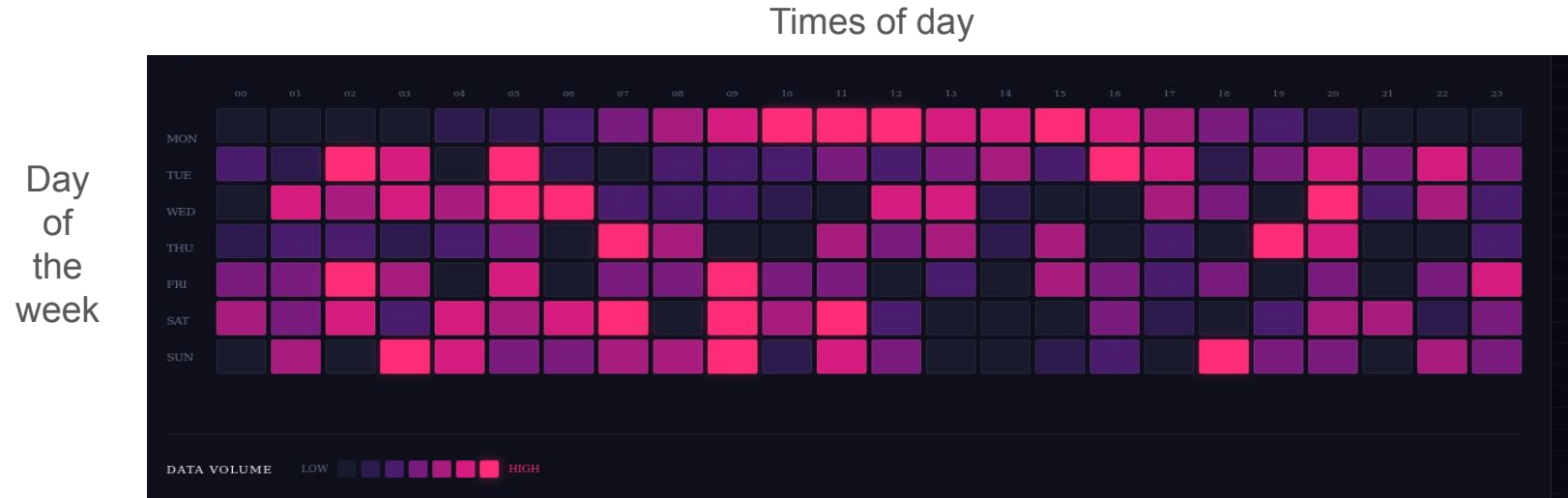
IP origin : view the ratio alerts / requests

19



Heatmap regular activity

Other macroscopic scale : time dimension



Total activity during
the period of the cell

Heatmap regular activity

21

CyberSentinel SOC

DASHBOARD

IP ORIGIN

HABIT REQUEST FLUX

QUIZZ DIFF

BLIND TEST

SEARCH SYSTEM...



Size of aggregation

7 WEEK

30 MONTH

365 YEAR

? CUSTOM

SOC Operational requests activity

Weekly Behavioral Distribution



Date range

START DATE
JAN 01, 2025

END DATE
DEC 31, 2025

HISTORICAL ☒ AGGREGATION

ALERTS ☒ REQUESTS

Choose the firewall

FIREWALL SELECTION: FW0 (All)

Ratio = alerts / requests

HIGHLIGHTS HIGH-RATIO CELLS

MIN RATIO THRESHOLD

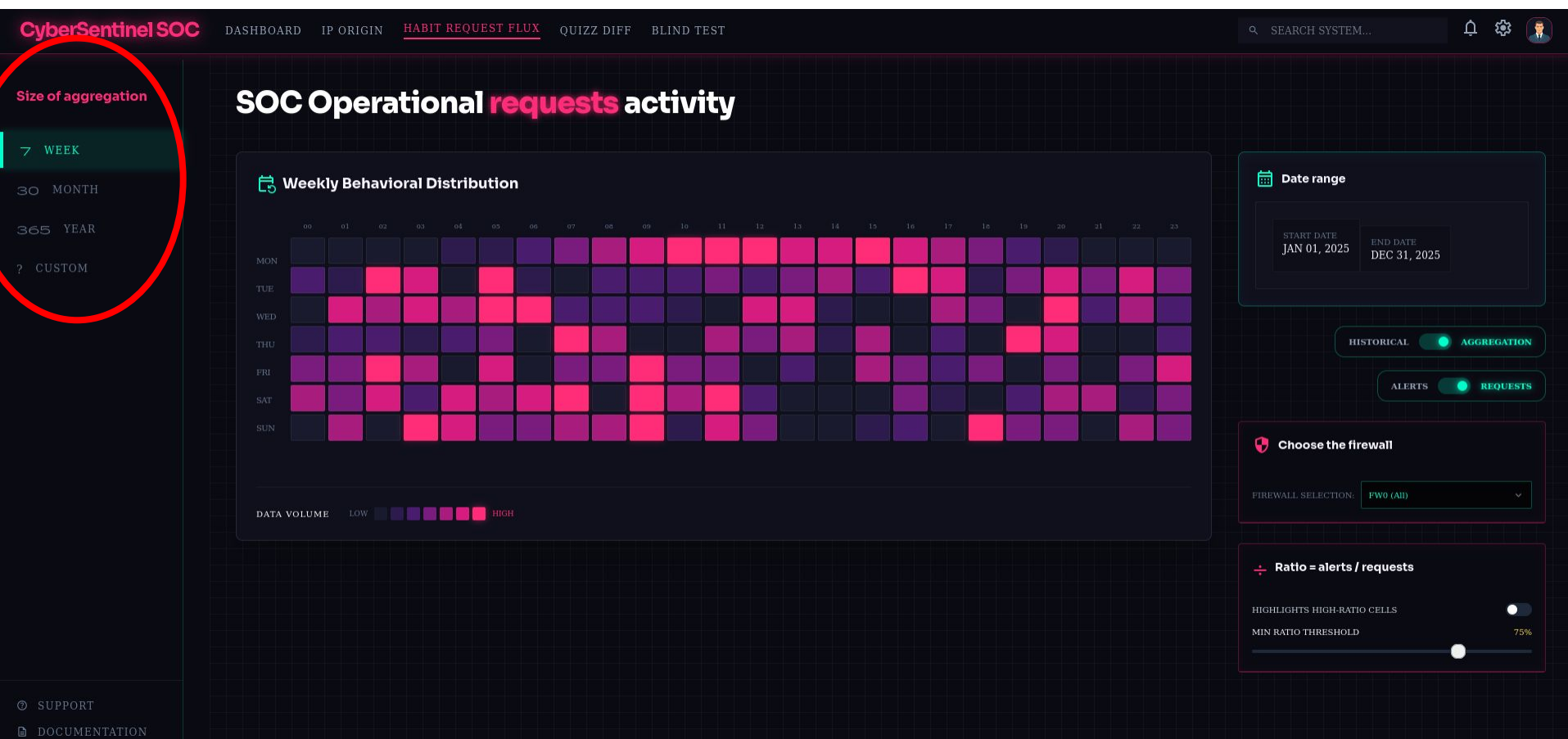
75%

SUPPORT

DOCUMENTATION

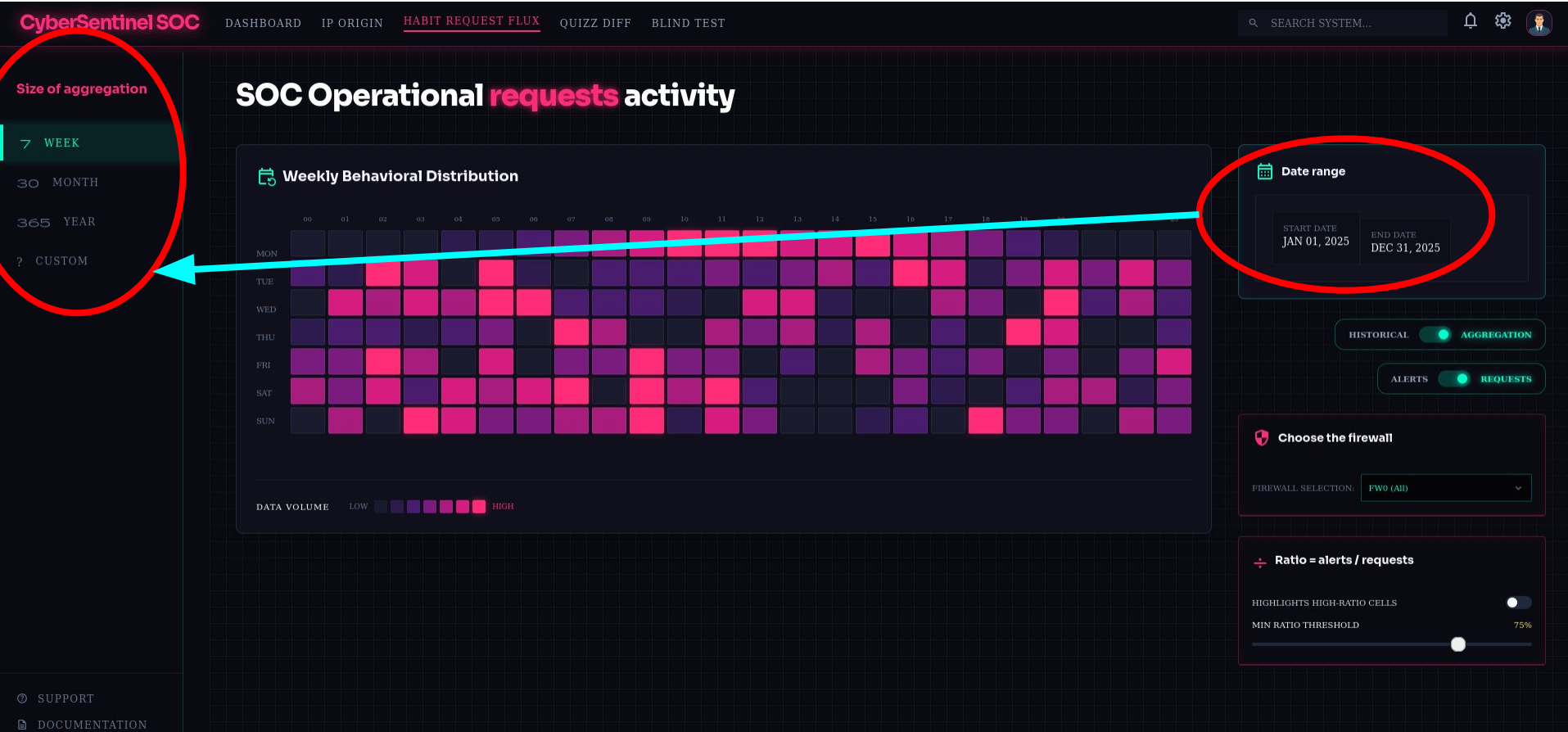
Heatmap regular activity : size of aggregation

22



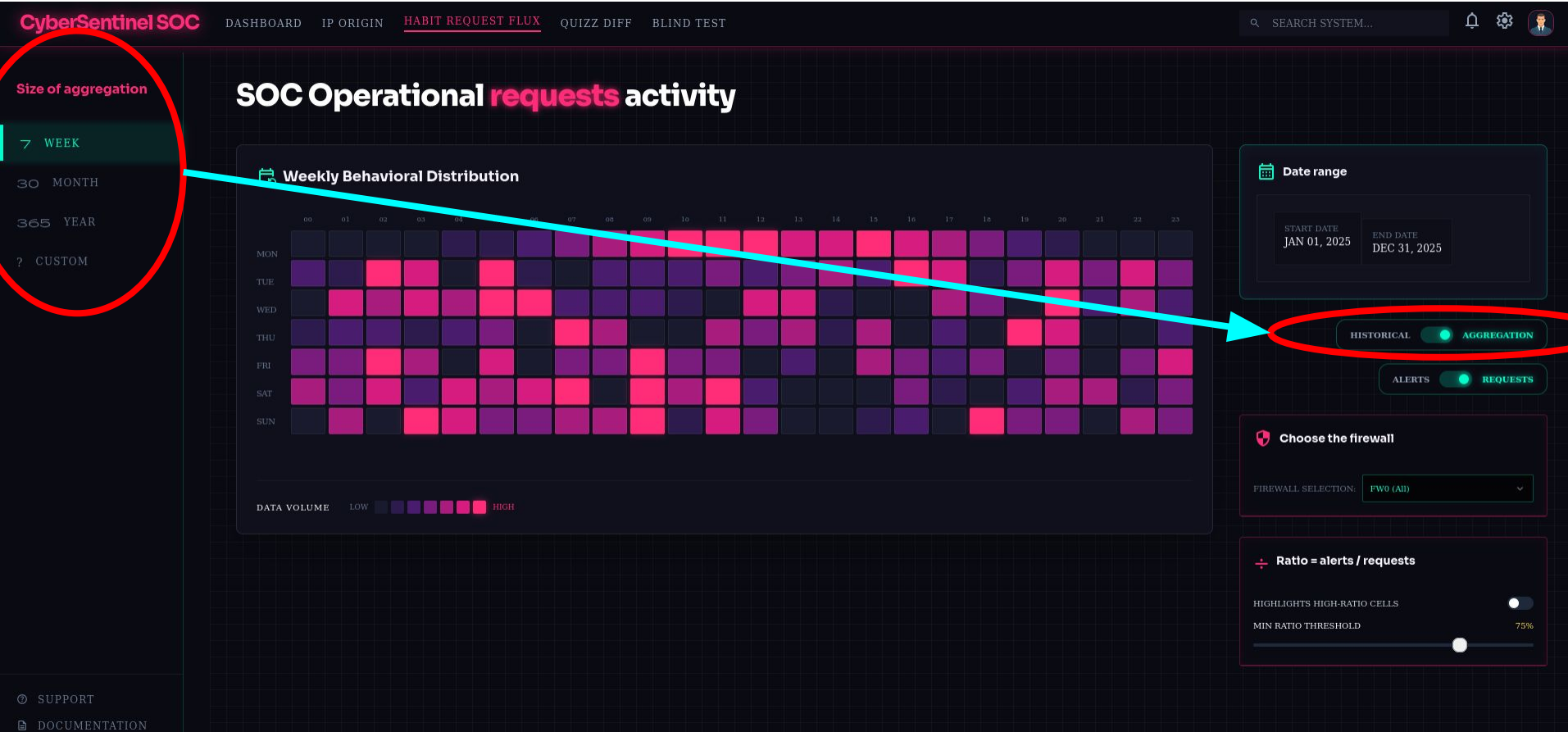
Heatmap regular activity : change range date

23



Heatmap regular activity : historical mod

24



Heatmap regular activity : choose firewall

25

CyberSentinel SOC

DASHBOARD

IP ORIGIN

HABIT REQUEST FLUX

QUIZZ DIFF

BLIND TEST

SEARCH SYSTEM...



Size of aggregation

7 WEEK

30 MONTH

365 YEAR

? CUSTOM

SOC Operational requests activity

Weekly Behavioral Distribution



Date range

START DATE
JAN 01, 2025

END DATE
DEC 31, 2025

HISTORICAL ☒ AGGREGATION

ALERTS ☒ REQUESTS

Choose the firewall

FIREWALL SELECTION: FW0 (All)

Ratio = alerts / requests

HIGHLIGHTS HIGH-RATIO CELLS

MIN RATIO THRESHOLD

75%

SUPPORT

DOCUMENTATION

Heatmap regular activity : request mod

26

CyberSentinel SOC

DASHBOARD

IP ORIGIN

HABIT REQUEST FLUX

QUIZZ DIFF

BLIND TEST

SEARCH SYSTEM...



Size of aggregation

7 WEEK

30 MONTH

365 YEAR

? CUSTOM

SOC Operational requests activity

Weekly Behavioral Distribution



DATA VOLUME LOW HIGH

Date range

START DATE
JAN 01, 2025

END DATE
DEC 31, 2025

HISTORICAL ☒ AGGREGATION

ALERTS ☒ REQUESTS

Choose the firewall

FIREWALL SELECTION: FW0 (All)

Ratio = alerts / requests

HIGHLIGHTS HIGH-RATIO CELLS

MIN RATIO THRESHOLD

75%

SUPPORT

DOCUMENTATION

Heatmap regular activity : alerts mod

27

CyberSentinel SOC

DASHBOARD

IP ORIGIN

HABIT REQUEST FLUX

QUIZZ DIFF

BLIND TEST

SEARCH SYSTEM...



Size of aggregation

7 WEEK

30 MONTH

365 YEAR

? CUSTOM

SOC Operational alerts activity

Weekly Behavioral Distribution



Date range

START DATE
JAN 01, 2025

END DATE
DEC 31, 2025

HISTORICAL ☒ AGGREGATION

ALERTS ☒ REQUESTS

Choose the firewall

FIREWALL SELECTION: FW0 (All)

Ratio = alerts / requests

HIGHLIGHTS HIGH-RATIO CELLS

MIN RATIO THRESHOLD

75%

SUPPORT

DOCUMENTATION

Heatmap regular activity : alert-to-request ratio

28

CyberSentinel SOC

DASHBOARD

IP ORIGIN

HABIT REQUEST FLUX

QUIZZ DIFF

BLIND TEST

SEARCH SYSTEM...



Size of aggregation

7 WEEK

30 MONTH

365 YEAR

? CUSTOM

SOC Operational requests activity

Weekly Behavioral Distribution



Date range

START DATE
JAN 01, 2025

END DATE
DEC 31, 2025

HISTORICAL ☒ AGGREGATION

ALERTS ☒ REQUESTS

Choose the firewall

FIREWALL SELECTION: FW0 (All)

Ratio = alerts / requests

HIGHLIGHTS HIGH-RATIO CELLS

MIN RATIO THRESHOLD

75%

Heatmap regular activity : show ratio

29

CyberSentinel SOC

DASHBOARD

IP ORIGIN

HABIT REQUEST FLUX

QUIZZ DIFF

BLIND TEST

SEARCH SYSTEM...



Size of aggregation

7 WEEK

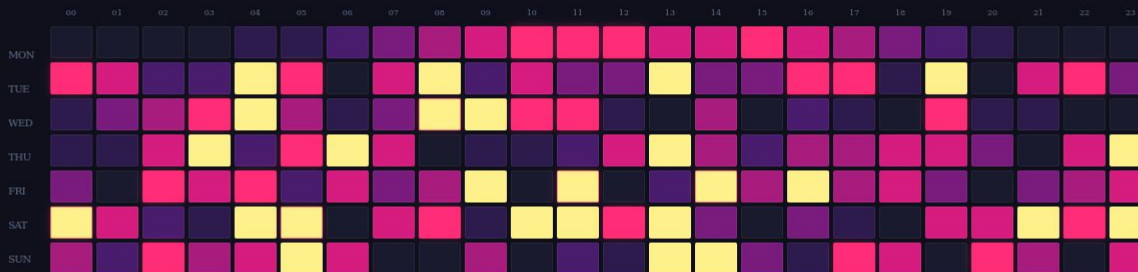
30 MONTH

365 YEAR

? CUSTOM

SOC Operational requests activity

Weekly Behavioral Distribution



DATA VOLUME LOW HIGH

Date range

START DATE
JAN 01, 2025

END DATE
DEC 31, 2025

HISTORICAL ☒ AGGREGATION

ALERTS ☒ REQUESTS

Choose the firewall

FIREWALL SELECTION: FW0 (All)

Ratio = alerts / requests

HIGHLIGHTS HIGH-RATIO CELLS

MIN RATIO THRESHOLD

75%

SUPPORT

DOCUMENTATION

Before and after comparison form : question

30

SOC comparative

Your olds habits

OUTILS (SIEM, EDR...)

Write your answer

OUTILS (SIEM, EDR...)

Write your answer

OUTILS (SIEM, EDR...)

Write your answer

OUTILS (SIEM, EDR...)

Write your answer

OUTILS (SIEM, EDR...)

Write your answer

CONFIRM MY HABITS

Before and after comparison form : question

31

CyberSentinel SOC

DASHBOARD

IP ORIGIN

HABIT REQUEST FLUX

QUIZZ DIFF

BLIND TEST

SEARCH SYSTEM...



SOC comparative

Your olds habits

OUTILS (SIEM, EDR...)

...

OUTILS (SIEM, EDR...)

...

OUTILS (SIEM, EDR...)

...

OUTILS (SIEM, EDR...)

...

OUTILS (SIEM, EDR...)

...

CONFIRM MY HABITS

Before and after comparison form : answer

32



SOC comparative

Your olds habits

OUTILS (SIEM, EDR...)

Write your answer

↗ here : ...

OUTILS (SIEM, EDR...)

Write your answer

↘ here : ...

OUTILS (SIEM, EDR...)

Write your answer

↘ here : ...

OUTILS (SIEM, EDR...)

Write your answer

→ here : ...

OUTILS (SIEM, EDR...)

Write your answer

↘ here : ...

CONFIRM MY HABITS

PATTERN HISTORY



● ACCEPTABLE ACTIVITY
ID: #41 - 04:22 UTC



● CRITICAL ALERT
ID: #40 - 04:15 UTC



● ACCEPTABLE ACTIVITY
ID: #39 - 04:02 UTC



● CRITICAL ALERT
ID: #38 - 03:50 UTC

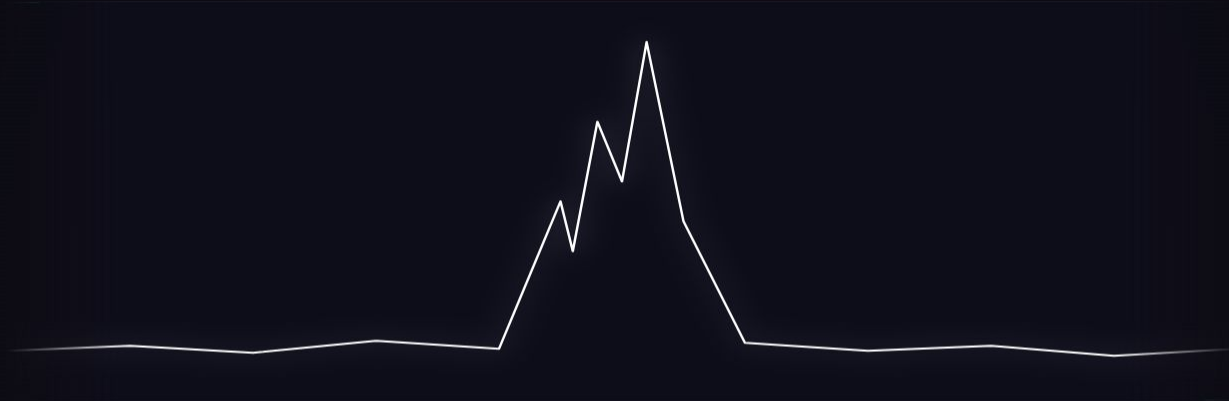
LOAD MORE DATA

🔗 SUPPORT

📄 DOCUMENTATION

Pattern Recognition #42

ANALYZE ACTIVITY CURVE FOR POTENTIAL LATERAL ESCALATION



CRITICAL ALERT



ACCEPTABLE ACTIVITY



HINT



Blind test : pattern recognition

34

CyberSentinel SOC

DASHBOARD

IP ORIGIN

HABIT REQUEST FLUX

QUIZZ DIFF

BLIND TEST

SEARCH SYSTEM...



PATTERN HISTORY



● ACCEPTABLE ACTIVITY

ID: #41 - 04:22 UTC



● CRITICAL ALERT

ID: #40 - 04:15 UTC



● ACCEPTABLE ACTIVITY

ID: #39 - 04:02 UTC



● CRITICAL ALERT

ID: #38 - 03:50 UTC

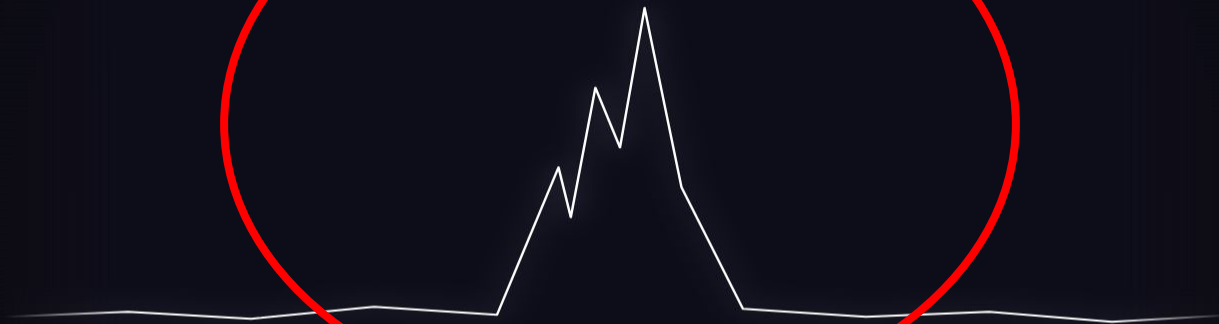
LOAD MORE DATA

🔗 SUPPORT

📄 DOCUMENTATION

Pattern Recognition #42

ANALYZE ACTIVITY CURVE FOR POTENTIAL LATERAL ESCALATION



CRITICAL ALERT



ACCEPTABLE ACTIVITY



HINT



Blind test : hint

35

CyberSentinel SOC

DASHBOARD

IP ORIGIN

HABIT REQUEST FLUX

QUIZZ DIFF

BLIND TEST

SEARCH SYSTEM...



PATTERN HISTORY



● ACCEPTABLE ACTIVITY

ID: #41 - 04:22 UTC



● CRITICAL ALERT

ID: #40 - 04:15 UTC



● ACCEPTABLE ACTIVITY

ID: #39 - 04:02 UTC



● CRITICAL ALERT

ID: #38 - 03:50 UTC

LOAD MORE DATA

🔗 SUPPORT

📄 DOCUMENTATION

Pattern Recognition #42

ANALYZE ACTIVITY CURVE FOR POTENTIAL LATERAL ESCALATION



CRITICAL ALERT



ACCEPTABLE ACTIVITY



HINT



PATTERN HISTORY

Pattern Recognition #42

ANALYZE ACTIVITY CURVE FOR POTENTIAL LATERAL ESCALATION



TACTICAL INSIGHT

Correct choice logic: This specific spike profile shows a multi-stage escalation typical of lateral movement. Note the secondary "echo" peaks at 550ms which differentiate it from standard automated backups that maintain a flat saturation plateau.

● ACCEPTABLE ACTIVITY
ID: #39 - 04:02 UTC

● CRITICAL ALERT
ID: #38 - 03:50 UTC

LOAD MORE DATA

🔗 SUPPORT

📄 DOCUMENTATION

CRITICAL ALERT



ACCEPTABLE ACTIVITY



HINT

Blind test : correction

37

CyberSentinel SOC

DASHBOARD

IP ORIGIN

HABIT REQUEST FLUX

QUIZZ DIFF

BLIND TEST

SEARCH SYSTEM...



PATTERN HISTORY



● ACCEPTABLE ACTIVITY
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ID: #40 - 04:15 UTC



● ACCEPTABLE ACTIVITY
ID: #39 - 04:02 UTC



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ID: #38 - 03:50 UTC

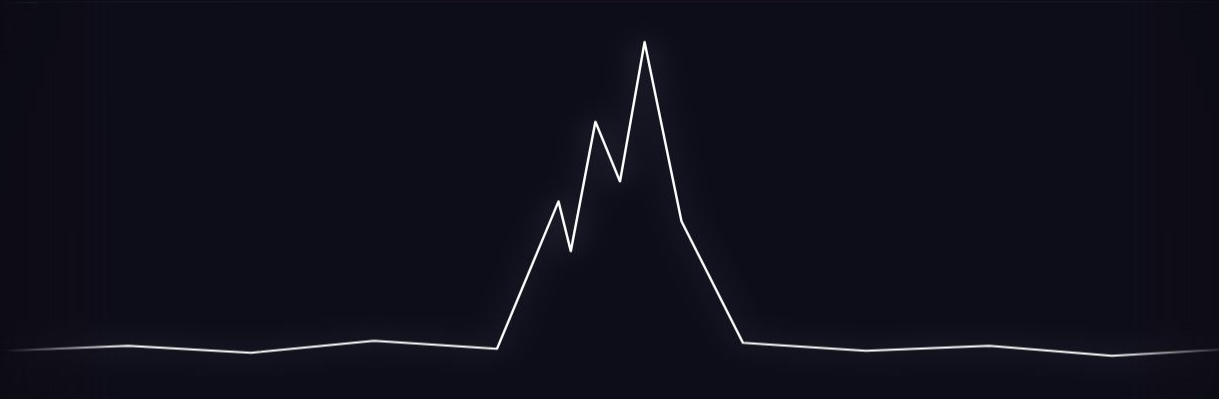
LOAD MORE DATA

🔗 SUPPORT

📖 DOCUMENTATION

Pattern Recognition #42

ANALYZE ACTIVITY CURVE FOR POTENTIAL LATERAL ESCALATION



CRITICAL ALERT



ACCEPTABLE ACTIVITY

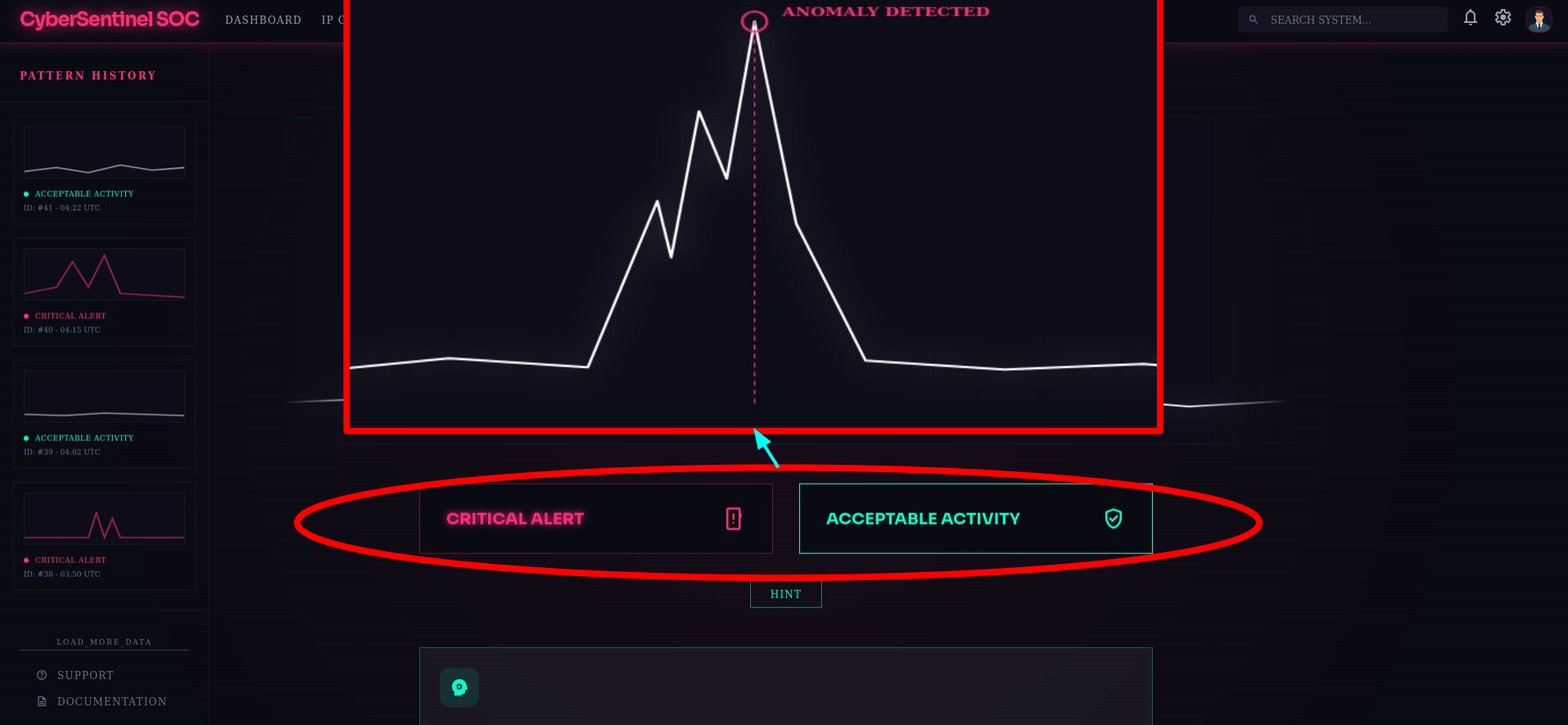


HINT



Blind test : correction

38



Blind test : history

39

CyberSentinel SOC

DASHBOARD

IP ORIGIN

HABIT REQUEST FLUX

QUIZZ DIFF

BLIND TEST

SEARCH SYSTEM...



PATTERN HISTORY



● ACCEPTABLE ACTIVITY

ID: #41 - 04:22 UTC



● CRITICAL ALERT

ID: #40 - 04:15 UTC



● ACCEPTABLE ACTIVITY

ID: #39 - 04:02 UTC



● CRITICAL ALERT

ID: #38 - 03:50 UTC

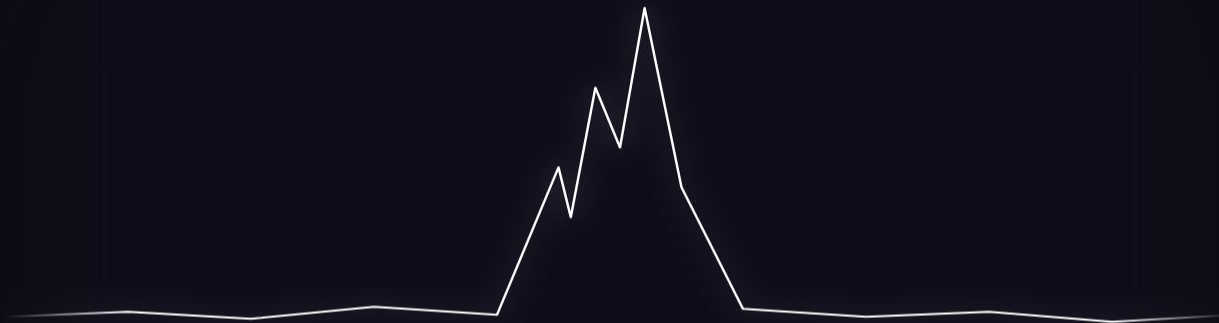
LOAD MORE DATA

SUPPORT

DOCUMENTATION

Pattern Recognition #42

ANALYZE ACTIVITY CURVE FOR POTENTIAL LATERAL ESCALATION



CRITICAL ALERT



ACCEPTABLE ACTIVITY



HINT



The end

Thanks you for your attention

Do you have any questions ?